Transcript

Week 28: Speech Recognition

Video 1 – Speech Recognition

Ever wondered how your phone understands your voice—or how virtual assistants respond so naturally?

Welcome to our session on Speech Recognition. Let's begin this exciting journey into the world where sound becomes data and speech becomes intelligence.

Let's take a quick look at what we'll cover in this module. First, we'll work to demystify speech recognition—understanding what it really means for machines to process human speech. Then, we'll take a journey through time, tracing the history of this technology and the breakthroughs that shaped it. Next, we'll go beyond the basics to explore why speech recognition is such an important enabler in modern life. Finally, we'll look at its real-world impact across industries, from healthcare to consumer technology.

Our primary goal is to demystify speech recognition. You'll see how machines process the sounds of our voice and convert them into text. We'll also look at key components such as acoustic modelling, feature extraction, and language modelling. Then, we'll take a journey through time—starting with early systems that could only recognise a few words, and moving forward to intelligent assistants like Siri and Alexa. You'll see how breakthroughs and computing power reshaped the field.

Next, we'll go beyond the basics. Speech recognition is a pivotal technology in our modern world, and it plays a vital role across many sectors. Finally, we'll explore its real-world impact. You'll see how it is being used in healthcare, in customer service, and in consumer technology, powering the smart assistants and devices we use every day. By the end, you'll understand what speech recognition is, how it evolved, why it matters, and how it is shaping the way we live and interact with technology.

Are you ready to find out how your voice becomes data—and how that data powers everything from smart assistants to healthcare tech? Let's get started.

Video 2 - Basics of Speech Recognition – Introduction

Have you ever spoken to your phone and wondered how it knows what you're saying? Speech recognition is the technology that makes that possible. In this session, we'll break down the basics and see why understanding this process is essential for anyone studying modern A-I. Let's explore how machines turn sound into meaning.

In this session, we'll uncover the fascinating process behind how machines, like your phone or smart speaker, understand what you say. Speech recognition may seem like magic—but it's actually a powerful blend of algorithms and data. You'll learn how your voice is captured through a microphone, then turned into digital signals that computers can process. By the end, you'll understand the essential steps that make voice-based technologies work—from input to output.

Let's begin by clearly defining what we mean by speech recognition. At its core, it's a branch of artificial intelligence that enables computers to process human language. Think of it as teaching a computer to listen to speech and then write down what it hears in digital text. You'll often come across terms like Automatic Speech Recognition, or A-S-R, and Speech-to-Text, or S-T-T. These names describe the same process: taking audio input and converting it into machine-readable text. The ultimate goal is to bridge the gap between how humans communicate and the structured language that computers can process and act upon.

Now, how does this process actually work? While the underlying algorithms can be complex, we can break it down into six main stages — a journey from raw sound waves to meaningful text.

It all begins with Speech Capture. This is where your voice is recorded through a microphone, converting the sound waves into a digital format that a computer can work with.

Once that's done, we move to Audio Processing. Here, the system cleans up the raw recording — removing background noise, adjusting the volume, and preparing the signal so that it's easier to analyse.

Next, we have Feature Extraction. Just like how our ears and brains can pick out pitch, loudness, and emphasis in speech, machines need to identify these characteristics too. The system detects key spectral properties of the sound, which are crucial for telling one speech sound apart from another.

These features are then passed to an Acoustic Model. This model is trained to recognise phonemes — the smallest units of sound that distinguish one word from another. For example, the kuh sound in the word cat. The model learns to map these sound patterns to specific phonemes.

At the same time, a Language Model comes into play. Human speech is not just random sounds — it follows rules and patterns. The language model predicts which word sequences are most likely to occur in context. For example, it knows that the phrase recognise speech is far more likely than wreck a nice peach, even though they sound similar.

Finally, we reach Decoding and Transcription. This is where everything comes together. The system combines the probabilities from the acoustic model and the predictions from the language model to generate the most likely sequence of words. The result is the final written version of what you said.

So, from capturing your voice to producing accurate text, this six-stage process is what makes modern speech recognition work so seamlessly."

Let's take a closer look at the first major step in speech recognition: the microphone and preprocessing stage, also known as the listener. Your voice is an analogue signal — it flows continuously and varies in pitch and loudness. The microphone captures that signal, converting it into electrical signals.

Once the signal is captured, audio preprocessing begins. The first and most critical step is analogue-to-digital conversion, or A-D-C. This process samples the electrical signal at regular intervals, turning it into discrete digital values that a computer can understand. You can imagine this as creating tiny snapshots of sound over time.

Next comes noise reduction. This filters out unwanted background sounds, like humming fans or keyboard clicks, so the system can focus on your voice.
Then we have normalisation, which ensures the volume is consistent no matter if you speak softly or loudly.

Finally, the audio is broken down into frames and windows. Instead of analysing the entire signal at once, it's divided into overlapping segments, each lasting about twenty to thirty milliseconds. This allows the system to analyse short, stable chunks of speech, which is crucial because speech can change very quickly.

Altogether, this stage transforms raw sound into clean, usable data that the system can process in the next steps of recognition.

Once we've cleaned, digitised, and framed the audio, the next crucial step is feature extraction. This is where raw audio is transformed into a more compact and meaningful numerical representation that a computer can understand.

Think of it like this. If you were trying to identify different musical instruments by their sound, you wouldn't just stare at the raw waveform. Instead, you'd pay attention to the qualities that make them unique — their timbre, their pitch, and how loud they are. Feature extraction does something similar for speech.

One of the most widely used techniques is called Mel-Frequency Cepstral Coefficients, or M-F-C-Cs. These are designed to mimic how the human ear perceives different frequencies, especially the ones most important for speech. They capture the timbre, or tonal quality, of your voice.

Other important features often extracted include pitch, which refers to the fundamental frequency and gives your voice its perceived highness or lowness, and energy, or loudness, which reflects the strength or intensity of the sound.
We extract these features because they significantly reduce the complexity of the data while keeping the crucial details intact. This prepares the data for the next stage — the recognition models that interpret speech accurately.

Now we reach the brains of speech recognition, where two models work together. The first is the acoustic model, which acts like the system's ears. It is trained on large collections of speech recordings with transcripts, learning how sound features match the smallest units of language called phonemes. Based on the features, it assigns probabilities — for example, an eighty percent chance the sound is an ah, and a twenty percent chance it's an eh. In this way, the acoustic model maps audio patterns to phonemes, much like ears recognising sounds.

Alongside the acoustic model is the language model, which adds meaning and context. Instead of focusing on sounds, it is trained on massive text corpora — books, articles, and transcripts — to learn grammar, vocabulary, and common patterns of language. It predicts which word sequences are most likely. For example, it knows "recognise speech" is highly probable, while "wreck a nice peach" is not, even if they sound alike.
Together, the two models ensure the system produces transcriptions that are accurate in both sound and meaning.

We've now reached the final stage of speech recognition, known as the decoder, or search algorithm. This is where the system brings everything together. Its role is to combine the information from both the acoustic model and the language model and decide on the most likely sequence of words.

The decoder explores many possible word sequences. For each option, it assigns a score, based on two factors — how well the sounds match the predicted words from the acoustic model, and how probable that sequence is in the language model. By weighing these probabilities, the decoder finds the best path — the sequence with the highest overall likelihood

Once the decoder identifies this best path, it outputs the final transcribed text — what you actually see when speech is turned into written words on your device. You can think of the decoder as the problem solver of the system, making sense of all the information. And what's truly impressive is that this entire decision-making process — from sound capture to readable text — takes place in just a few milliseconds.
Can you map each step of the speech recognition pipeline—from microphone input to decoded text—using a real-life voice assistant example? Try it, and see if you can identify where each component plays its part.

Video 3 - A Journey Through Time: The History of Speech Recognition

Think of the last time you used Siri, Alexa, or Google Assistant. Ever wondered what paved the way for them? The story starts decades ago

Speech recognition starts with capturing sound. Microphones first convert sound waves into signals. These signals are then turned into digital samples. Background noises such as traffic or fan hums are filtered out, and the audio is adjusted to a consistent level for accuracy. The final step is framing and windowing. Here, the continuous audio stream is divided into small, overlapping segments, known as frames. This makes the data easier for the system to analyse and recognise patterns in speech.

Our story begins in 1952 with Bell Laboratories' system, Audrey. This was groundbreaking as the first machine able to recognise spoken digits from zero to nine. But it had big limitations. Audrey was speaker-dependent, meaning it only worked reliably with the developer's own voice. It could recognise individual words but not continuous speech, so users had to speak carefully and clearly.

A decade later, in 1962, IBM introduced Shoebox. It could understand 16 English words, including digits and arithmetic commands like plus and total. You could even solve sums by speaking to it. Yet Shoebox faced the same hurdles as Audrey. It was still speaker-dependent, needed deliberate pauses between words, and had a very limited vocabulary.

The 1970s marked a turning point in speech recognition. The DARPA Speech Understanding Research Program poured resources into ambitious projects. One major outcome was Carnegie Mellon University's Harpy system in 1976. Harpy recognised more than 1,000 words in constrained domains. Compared to earlier systemswhich was limited to a few dozen words, this was a huge breakthrough, showing the power of statistical models over rigid, rule-based systems.

In the 1980s, speech recognition advanced with Hidden Markov Models, or HMMs. These models used probabilities to capture speech patterns. They allowed systems to move towards speaker independence and, most importantly, enabled continuous speech recognition without deliberate pauses. A major example was IBM's Tangora system in the mid-1980s. Tangora was a voice-activated typewriter with a vocabulary of over 20,000 words, demonstrating how statistical models greatly improved the power and usability of speech recognition.

In 1990, Dragon Systems launched Dragon Dictate, the first widely available consumer speech recognition product. It was a major step, but it came at a high price and required hours of training to adapt to each user's voice. It also still relied on discrete speech. A real breakthrough came in 1997 with Dragon Naturally-Speaking. This was the first system to offer continuous speech recognition for consumers, allowing natural speech without pauses. It could transcribe at speeds of around 100 words per minute, powered by the increasing processing power of personal computers.

By the early 2000s, speech recognition started to appear in more practical, everyday uses. Telephony systems, known as Interactive Voice Response or I-V-R, let callers navigate menus using spoken commands. Early mobile voice commands also emerged, though they were often clunky. Despite progress, challenges remained. Error rates were still high, systems needed a lot of manual correction, and applications beyond dictation were limited.

The 2010s marked a turning point for speech recognition with the rise of deep learning. Advances in neural networks, supported by larger datasets and powerful GPUs, allowed systems to capture complex speech patterns. Models like recurrent and convolutional neural networks drastically reduced the Word Error Rate, making recognition far more accurate than older HMM-based systems.

With these breakthroughs, speech recognition moved from a niche technology to a mainstream feature. In 2011, Apple introduced Siri, putting an intelligent voice assistant into millions of smartphones. This was followed by Amazon Alexa in 2014, which popularised smart speakers in homes. In 2016, Google Assistant brought highly accurate voice capabilities across devices. Today, modern systems can reach near-human accuracy, handle multiple languages and accents, and even achieve real-time transcription. Open-source models, like OpenAI's Whisper, continue to expand accessibility and push the boundaries of speech recognition.

Speech recognition once relied on modular systems, where separate components such as acoustic and language models worked together in a pipeline. With end-to-end deep learning, this shifted to a unified approach. A single large neural network now performs direct mapping from raw audio to text, replacing the need for multiple specialised models.

The advantages of end-to-end deep learning are profound. Training is significantly simplified. Instead of having to optimise multiple separate models, researchers only need to train one large network. Another benefit is reduced error propagation. In modular systems, errors from one stage could cascade into others. End-to-end systems avoid this by learning the entire mapping internally. Most importantly, they provide richer contextual understanding by capturing relationships directly from audio, making them robust to accents, speaking styles, and environmental noise. These benefits were demonstrated in pioneering models such as Baidu's Deep Speech and Google's Listen, Attend and Spell, which proved the power of this unified approach.

In 2017, a team at Google Brain introduced Transformers in their paper *Attention Is All You Need*. While first designed for language tasks, Transformers soon found their way into speech. Their core innovation is the self-attention mechanism. Unlike earlier models that processed input step by step, Transformers could look at an entire sequence at once, learning which parts of the input were most important.
Transformers proved incredibly powerful for speech. They excel at capturing long-range dependencies — for example, linking the start of a sentence with a word much later on. They also enable massive parallel processing, handling large chunks of audio at once instead of step by step. This allowed faster training and made use of much larger datasets. With this efficiency, Transformers replaced recurrent networks in many speech models and became the new standard for handling long audio sequences.

One of the biggest breakthroughs in modern speech recognition is Wav2Vec 2.0, developed at Facebook AI Research, now Meta AI, by Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. Its unique power comes from self-supervised learning, or SSL. Instead of relying on transcribed audio, the model learns directly from raw, unlabelled audio data. It masks out parts of the input and predicts them, similar to how the BERT model works for text. It predicts the missing parts based on the surrounding context. By repeating this across vast amounts of untranscribed audio, the model develops incredibly rich and robust internal representations of speech.

The impact of Wav2Vec 2.0 is immense. It significantly reduces the need for vast amounts of transcribed audio, which is costly and time-consuming to produce. This makes it possible to build high-performance systems even for low-resource languages, where labelled data is limited. A model can be pre-trained on large amounts of unlabelled audio and then fine-tuned on just a small labelled set. Its architecture is based on a powerful Transformer encoder, giving it robust and accurate representations of speech.

One of the most important modern speech recognition models is the Conformer, developed by researchers at Microsoft, including Ananya Chen, Cheng Gong, Yu Wu, and Jinyu Li. What makes Conformer unique is its hybrid architecture. It combines Transformers, which excel at capturing global context and long-range dependencies, with Convolutional Neural Networks, which are strong at identifying local, fine-grained features such as phonemes and subtle variations. This powerful combination allows the Conformer to capture both global and local aspects of speech.

The Conformer integrates Transformers and CNNs within specialised Conformer blocks, allowing it to process both broad linguistic context and precise acoustic details of speech at the same time. This balanced design enables the model to achieve state-of-the-art results across speech recognition benchmarks. By effectively handling both acoustic and linguistic features in a unified framework, Conformer offers a highly accurate and robust solution for complex automatic speech recognition tasks.

The next frontier in speech recognition is the rise of Speech Large Language Models, or Speech LLMs. These adapt large models like GPT or Llama to directly process and understand speech. Their greatest strength is contextual understanding, using knowledge from text training to capture nuance, disambiguate homophones, and infer meaning beyond literal transcription. They also extend into generative capabilities, such as summarising spoken conversations, translating languages in real time, or answering complex questions. Looking further ahead, Speech LLMs aim for multimodal reasoning, where speech can be combined with images or video for deeper understanding. Early progress can be seen in OpenAI's Whisper, a highly robust and multilingual ASR model. Current research is moving towards direct speech-to-language systems that map audio straight to language outputs without explicit transcription, marking a major leap forward in intelligent speech recognition.

How do you imagine Speech LLMs shaping the way we communicate, learn, or work in the next decade?

Video 4 - Why Speech Matters: The Importance of Speech Recognition

From the earliest experiments to today's powerful systems, speech recognition has travelled an incredible path — and the journey is still unfolding.

Now it's time to see why this technology is more than just a novelty. It has become an indispensable force reshaping industries and profoundly influencing our daily lives. In this session, we'll reveal role in daily lives and show impact on modern world.

One of the most profound impacts of speech recognition lies in enhancing accessibility and fostering inclusivity. This technology is transforming the lives of millions by enabling them to interact with devices in ways that might otherwise be difficult.

For individuals with motor impairments, it provides hands-free control of technology, allowing them to dictate emails, browse online, or operate smart homes purely through voice commands.

For those with visual impairments, it creates a voice-driven interface, making it possible to navigate, access information, and engage with applications without relying on sight.

Even for some with speech impairments, advanced systems are being designed to adapt to atypical speech patterns, offering augmented communication support and improving digital interactions.

Ultimately, speech recognition helps democratise access to technology, granting independence, better communication, and broader participation in the digital world. Beyond accessibility, speech recognition also drives productivity and efficiency across industries. Time is a valuable resource, and voice interaction helps save it.

In professional fields like medicine and law, it enables faster data entry, where dictation software can turn lengthy reports or notes into text in seconds instead of minutes.

Another major advantage is hands-free operation. Drivers can use voice commands for navigation or calls while keeping their focus on the road, and in industrial or lab settings, workers can log data or follow instructions without putting down tools.

Even at home, speech recognition provides everyday convenience, from setting timers while cooking to adding items to a shopping list, all without touching a screen.

By cutting down on typing and manual actions, speech recognition helps reduce effort and fatigue, allowing people to achieve more in less time

Customer service is an area that has been profoundly impacted by speech recognition technology. Gone are the days of frustrating, button-based Interactive Voice Response, or IVR, systems. Current IVR systems are far more sophisticated; instead of pressing buttons, you can simply speak your request naturally, and the system intelligently understands your intent to navigate you to the correct option, significantly reducing frustration and saving time.

The real payoff comes from tools like 'Agent Assist', which uses speech recognition to transcribe customer conversations in real-time. This provides the human agent with instant access to relevant information, policy details, or even pre-written responses. Furthermore, automated virtual agents and chatbots, powered by this same technology, are now handling a vast array of routine inquiries 24/7. This frees up human agents to focus on more complex, nuanced, or sensitive issues, leading to faster overall resolution times, reduced wait times for customers, and ultimately, enhanced customer satisfaction.

Speech recognition is not just a tool; it's the very voice of modern Artificial Intelligence, shaping how we interact with technology. It's the core interface for familiar smart speakers and voice assistants like Alexa, Google Assistant, and Siri, allowing you to get information, play music, or manage your home simply by speaking. This technology is also hugely prevalent in wearable technology. Your smartwatch or wireless headphones use voice control to enable seamless interaction and task management even when your hands are busy. Speech recognition also benefits I-o-T and home automation, allowing users to control everything from their lights and thermostats to their security systems with simple voice commands, and it's vital for hands-free control of navigation, music, and calls in modern cars, enhancing both convenience and safety.

Looking to the future, speech recognition is a foundational technology for a host of emerging AI applications. This includes more sophisticated conversational AI, where systems can maintain extended, natural dialogues. We'll also see advances in sentiment analysis from voice, which can detect emotional cues; voice biometrics for secure authentication; and increasingly complex multi-modal AI systems that combine speech with other inputs, such as vision, for a richer understanding of the world. Speech recognition is truly at the forefront of how we will interact with intelligent systems in the years to come.

Finally, let’s consider the broader economic and societal impact of speech recognition technology, which goes far beyond convenience.
Economically, it has driven job creation and new business models through research, development, deployment, and maintenance of ASR systems. It has also contributed to expanding the voice technology market rapidly, proving it is a robust and growing sector.
Societally, it is helping to break global communication barriers with translation, allowing people from different languages and cultures to interact more easily.
For businesses, it has enabled data-driven insights from spoken interactions, by converting speech into searchable text and unlocking valuable patterns.
Ultimately, it is helping to shape a more natural, accessible future, where human–computer interaction is intuitive, efficient, and empowering.

Now that we’ve explored how speech recognition shapes devices and future innovations, take a moment to reflect. How could you personally apply speech recognition to simplify or improve a task in your daily routine?

Video 5 - Speech in Action: Applications Across Domains

Think about the last time you spoke to Siri or Alexa — did you ever imagine the same technology could help surgeons, bankers, or even detectives?

Speech recognition isn't limited to personal assistants—it's transforming industries and daily life. Doctors can dictate notes straight into health records, drivers can navigate hands-free, call centre agents rely on live transcription, and captions make conversations accessible for everyone. Voice technology is no longer just a tool, it's a force shaping how we work, travel, and communicate.

Let's look at how speech recognition has moved from the lab into millions of homes, pockets, and cars. One of the most familiar examples is the smart speaker. Devices like the Amazon Echo with Alexa or Google Nest respond entirely to your voice. You can play music, set alarms, get news or weather updates, and even control smart lights or thermostats — all hands-free, with simple voice commands.

Then we have smartphone voice assistants, like Apple's Siri or Google Assistant. They are built into our phones and are deeply integrated into our daily routines. With just your voice, you can make calls, send texts, open apps, set reminders, or perform web searches — all without touching the screen.

Beyond phones, wearable technology such as smartwatches and wireless headphones is making voice control even more seamless. Quick commands allow you to answer calls, check notifications, or play music while you're on the move.

And finally, in the car, in-car infotainment systems have adopted voice control to make driving safer. You can use your voice to command navigation, change the music, adjust the climate, or even place a call — all while keeping your hands on the wheel and your eyes on the road.

Speech recognition is transforming the healthcare sector, where precision and efficiency matter most. One of its most significant uses is in clinical documentation and medical scribing. Doctors and nurses can dictate notes, diagnoses, and treatment plans directly into electronic health records. This not only reduces the burden of typing but also improves accuracy and allows clinicians to give more attention to their patients.

In telemedicine and virtual care, speech recognition is vital for transcribing online consultations, creating accurate records, and even supporting remote monitoring.

Even in highly sterile environments like the operating room, surgeons can use their voice to control equipment, adjust lighting, or access patient data, while maintaining sterility and focus on the procedure.

Finally, speech recognition improves accessibility for patients, giving those with mobility challenges the ability to control medical devices or communicate more effectively. In every case, voice technology is helping healthcare become more accurate, efficient, and human-centred.

In customer service and call centres, speech recognition is fundamentally changing how businesses interact with customers. Gone are the days of rigid systems where you had to press a number to get through. Now, modern IVR systems use natural speech to understand intent, letting customers simply say what they need, such as 'I want to dispute a charge,' and routing the call instantly.

For human agents, agent assist tools provide real-time transcription of calls and suggest responses based on knowledge articles or policy details. This not only saves time but also helps agents deliver more accurate and consistent service.

Meanwhile, voicebots and virtual assistants handle routine tasks like checking balances or answering frequently asked questions around the clock, freeing human agents to focus on complex or empathetic interactions.

And finally, through call analytics, businesses can analyse spoken data from thousands of conversations to identify customer pain points, evaluate performance, and continuously improve their services. Together, these applications show how voice is reshaping the customer service experience.

Speech recognition plays a vital role in transcription and captioning services, making spoken information easier to access and use.

Think about meetings and lectures. Instead of rushing to take notes, software can generate a transcript of everything that’s said. These transcripts form searchable archives, help with reviewing key points, and allow content creators to repurpose spoken words into written summaries or articles.

One of the most impactful uses is live captioning. Speech recognition creates real-time captions for television, online videos, and virtual meetings. This is essential for accessibility, helping people who are deaf or hard of hearing. It also benefits anyone in noisy environments, or learners who find it easier to follow along with text.

In journalism and content creation, voice-to-text speeds up the process of transcribing interviews, podcasts, and video footage. This gives journalists a quicker way to edit and analyse material.

Finally, in legal proceedings, speech recognition is used to assist court reporters by providing real-time transcripts. This improves efficiency and accuracy in handling legal documentation.

Speech recognition is also making a critical impact in security and biometrics, where your voice itself becomes a unique identifier.

One of the most powerful uses is voice authentication. Instead of relying on passwords or PINs, systems can recognise the unique characteristics of your voice, often called a ‘voiceprint’. This can secure access to bank accounts, devices, or even physical spaces, and it’s far harder to spoof than a simple password.

Beyond this, speech recognition supports fraud detection. By analysing speech patterns, systems can flag potential fraudulent callers or unusual activity, providing an extra layer of protection in customer interactions.

Finally, in more specialised settings, forensic voice analysis is used by law enforcement. Investigators can identify individuals from recorded speech, using advanced recognition and pattern-matching techniques to support criminal investigations.

In all of these areas, voice technology is reshaping how we protect identity and maintain security.

Now, let's explore some of the more specialised and emerging applications of speech recognition. This is where it goes beyond the obvious.

In education, it's a huge step forward for language learning, offering instant pronunciation feedback to help students refine their accents. It's also being used to power automated grading for spoken assignments and to create engaging, interactive learning experiences where students can converse with virtual tutors.

And in gaming, it enables a truly immersive experience, allowing you to control characters and interact with the game using your voice. The media and entertainment industry is using it for automated content indexing. This allows them to quickly search vast archives of film and TV footage for specific dialogue. For us, it’s behind the voice commands on streaming services, letting you find content just by speaking a title or keyword.

Finally, in robotics and smart environments, speech recognition is enabling a more natural interaction between humans and robots. You can speak commands to control robots in complex manufacturing settings or interact with service robots in public spaces. It's all about making control and communication more intuitive.

As you can see, speech recognition is a pervasive technology, constantly expanding its reach and impact. From enhancing daily tasks to driving cutting-edge innovations, its significance will only continue to grow."

The next exciting frontier in speech recognition is the convergence with Large Language Models, or LLMs. The current direction for research is to integrate these powerful LLMs directly into how we process and understand speech.

The promise of these Speech-Aware LLMs is immense. For a start, they bring unprecedented contextual understanding. Instead of just transcribing words, they can interpret the meaning beyond the literal transcription. This allows them to disambiguate homophones or understand implied meaning that goes beyond the literal words.

They also have powerful generative capabilities. These models don't just transcribe; they can automatically summarise spoken conversations, translate spoken language in real-time, or directly answer complex questions asked verbally.

Ultimately, they aim for multimodal reasoning. This is where they can understand spoken language in conjunction with other forms of media, like images or video, creating truly intelligent conversational agents.

A great example of a model that's already demonstrating aspects of this future is OpenAI's Whisper. While primarily an ASR model, Whisper is remarkable for its robustness and multilingual capabilities, hinting at the power of large models for speech. This truly represents the next generation of how machines will listen, understand, and interact with us.

Could speech-driven AI become the next leap in human–machine interaction — moving us from machines that only hear, to machines that truly understand, reason, and respond like humans?

Video 6- Speech Recognition- Demo

Hello and welcome everyone. Welcome to this short hands-on session where we will try to get comfortable with the first step of speech recognition that is we are going to load an audio, look at it in the time domain and then further peek into the frequency domain with some really nice looking spectrograms. So, that we are going to start with the Python library Librosa because it gives us simple powerful functions for audio analysis.

Just before we proceed ahead if you want I can actually help you to go through this Librosa documentation, a very well maintained library for the fundamental implementation of your speech recognition. So, you see it says that Librosa is a Python package for music and audio analysis. There are many more other libraries for advanced sort of you know speech recognition you have ESPNet.

So, you can explore that. It is more or less you know yeah. So, this is more on the bash side of implementation.

So, you would see that you know you still have some amount of shell scripting involved here although Python is 60.2 percent. So, you can use this also for advanced processing, but to keep it simple I would start with Librosa library. Now, you can again you can simply instal the Librosa library you have installation instructions over here.

This is how you can simply instal Librosa pip instal Librosa that is how and then we can get started with our next set of work. So, I have that command over here all the necessary libraries which are required we are able to instal them and then we are going to import the audio clip. So, we are going to use Librosa as you know for audio analysis, we are going to use numpy and matplotlib for all the maths and all the plotting and sound file we are going to do for input output audio.

Now, let us move to the next one once this finishes we need an audio clip. Typically a small wave file would be good someone speaking if you do not have one handy no problem there is always a sample wave file provided to you. So, this notebook can actually generate a simple dummy tone.

So, you can use that for the rest of the code still it runs. So, that is what we are doing we are going to import all these necessary libraries the only thing you need to change is the path to your audio file. So, this is the path which you want to use.

Now, if that file is not found the notebook will then you see it says that if the file is not found then the notebook is going to synthesise that file for you using some sampling rate and some you know technical details about that. So, well by the end of that you would see that it would print that dummy audio file and you know it would basically start with that audio file. But I would recommend you to use you know available wav file instead of this one because that would give you a more you know exciting relation with what we are going to see in the output right the waveforms.

Now, let us start the next obvious step is to load the file. So, when you load the file you do it with something called as librosa dot load and you provide the file path over here. Now, when we do that we get two things you see that you get a y and a sr y stands from the waveform samples and you would observe that it is a one dimensional float array.

Float array means it would be continuous numbers an array of continuous numbers and the other object which you tend to get is the sample rate in the form of hertz the samples per second. From these we can compute the duration and also you can play the clip right inside the notebook. So, for that we are going to just print these and you would see that the sample rate is printed, the audio signal is printed as I said it is going to be a one dimensional float array and also it prints you the duration around 8.33 seconds.

To play the sound right so, what you need to do is you need to just use the display method that is an audio player right. When you click play to quickly sanity check that the clip sounds right. So, you are going to display that you are going to pass your data and the sampling rate and it is going to show you this widget an inline audio player.

So, this confirms that we have read the real audio data and have the timing information and that is what we will need for plots. Now, the next obvious step is to look at the waveform in the time domain. You can think of it as amplitude versus time like looking at the raw vibrations.

And if you see this what are we doing here we are simply plotting the waveform right by passing my data that is my raw data and your sampling rate. My x axis is the time right you can see that and my y axis is amplitude you see this is my time and around 8.33 seconds is what the time you saw above. So, this is typically the same size in seconds around 8 seconds.

So, what do you see here? You see that for actual speech you will notice these bursts right you see these bursts right where syllables may happen and then you would see some quiet regions in between and at extremes right. So, you see this is at extreme and then again you see some quiet region here, some quiet region here, some dampen over here and then slowly it dies. So, this is a great for spotting clipping silences or any obvious glitches, but you clearly can see that it is not great for recognising words or what we call as phonemes.

For that we need frequency information. For a dummy tone you will see this is smooth periodic oscillation, but if you take a normal speech recorded you would see that you would see uneven bursts. So, as we learnt that you would see the bursts, but it is difficult for you to identify which words were spoken or which phonemes were spelled.

That is where we move ahead to the next type of analysis. We know that sound is not just how loud it is, it is also about which frequencies are present at each moment and that is what your spectrogram does. It would decompose the waveform into time cross frequency which is a heat map where you would see the colour would show the energy at a particular frequency and time.

Now, this is the bridge from your raw audio to features models can actually use. So, let us try to implement that. So, now, you know that we have a technique called as short time Fourier transform.

So, that is what you see STFT which is short time Fourier transform. It is a rolling fast Fourier transform over small windows and it would convert the magnitude to decibels for more interpretable scale. Again you would see that along the x axis is time.

So, you see that x axis is time and along the y axis you would see it is the frequency in hertz and then the colour would show the intensity. So, something like this. Now, for speech you will usually see horizontal bands right you would see that later we call that as formant.

So, the frequency regions that carry the vowel information. A brighter band indicates stronger frequency components right and that is what indicates what vowels, what syllables, what words, what phonemes are being spoken. So, now, you do not have only the bursts, but you also have the presence of frequencies at that time and the brighter the chart the more the present that frequency is and that is simply done by converting your amplitude to decibels and then you are just simply plotting those decibels with the sampling rate.

The spec show basically means that you are doing a spectrogram show. So, the reading tips for this spectrogram is you start from the left to right which is the time the frequency increases from bottom to top and brighter the colour means more energy. So, if you see a long horizontal line that would imply that they are steady tones.

Now, there is one more concept we call that as the Mel frequency or the Mel spectrogram and you know that humans do not hear all frequencies in a linear manner. There is something called as a Mel scale which compresses high frequencies and expands low ones to mimic our perception and this Mel spectrogram is the go-to starting point for many speech pipelines. You would see that you would take a you would later take logs and maybe deltas you are very close to features like log Mel spectrograms.

You would also see something called as MFCCs which are used in automatic speech recognition which you would see later. So, when I try to do a Mel spectrogram I just can change that to Mel spectrogram and everything else remains the same and you would see a slightly different which is perceptually motivated scales which often improves the recognition performance and reduces the feature dimensionality where our ears are very less sensitive. So, in the same manner where you had frequency now you have Mel frequency the structure is similar, but now it would emphasise lower frequency detail in a more detailed manner.

So, that is what we have done. The four foundational things we loaded an audio, we checked its properties, visualised the waveform, we computed the short time Fourier transform spectrogram and computed a Mel spectrogram. You can think of these steps as the front door to speech recognition.

Now, from here onwards you would typically see how do you create MFCCs and then there are voice activity detection and then you have ASR and TTS. Well, I would recommend you to try some of these experiments by changing the sample rate, by plotting the decibels, by trying a different tone and try to see if you can reproduce the same set of output and analyse from the waveform which you have uploaded does it actually mimic the waveform which you expect from the type of audio you have uploaded. Thank you everyone.